Document on cluster and cluster concepts:

1. What is cluster?

A cluster is a group of machines that works together as one.
Ex: 20 pods running apps

2. What is master node?

Master node is main of Kubernetes cluster and it controls all worker nodes.

3. What is worker node?

It runs pods and application containers.

4. What does master node contains?

API server: the api server is the main communication component of k8 it converts the request and returns responses.

ETCD: ETCD is a storage component of k8 cluster

Controller manager: the controller manager is responsible for maintaining the desired state of cluster. Like a pod stops running it takes action to fix the issue by creating new pod.

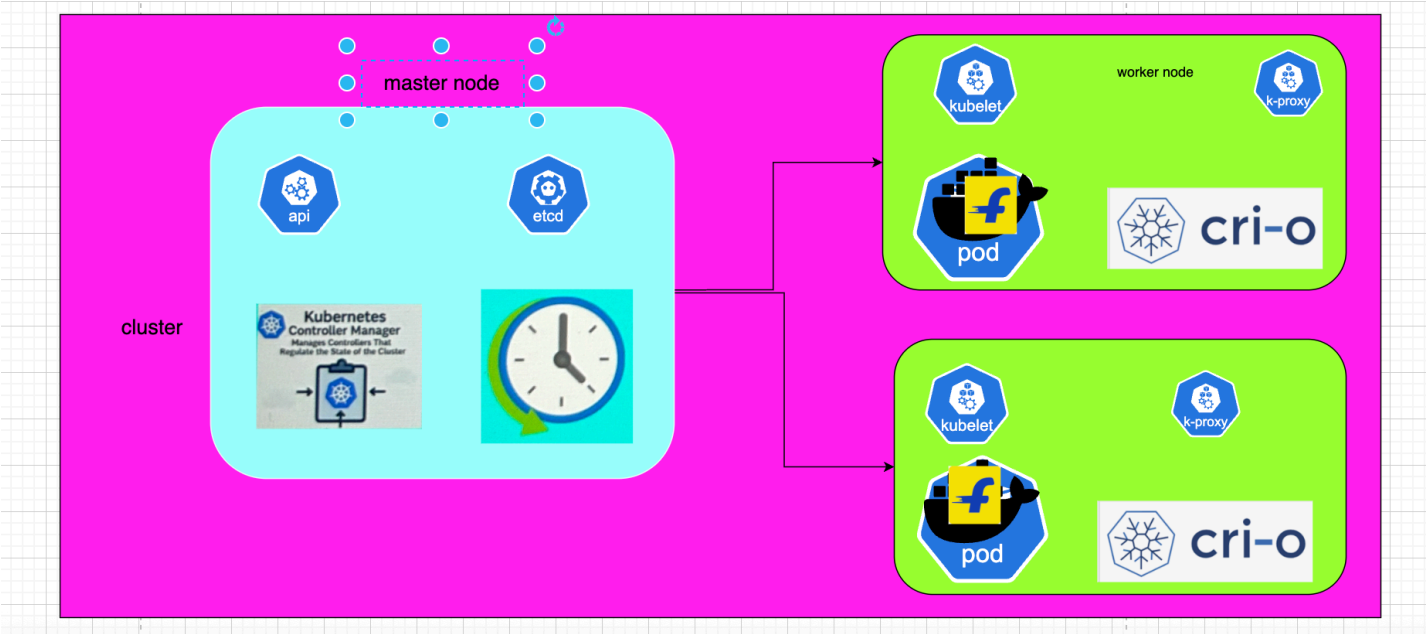
SCHEDULER: the schedule is used to assign the pods to worker nodes. It decided which worker node should run.

5. What does worker node contains?

KUBELET: kubelet is a communicates with the master node and follows its instructions. It checks the pods is running or not.

KUBE-PROXY: it handles the network communications and ensures the the traffic reaches the correct pod or not.

CONTAINER RUNTIME: it is a software used to run containers and it pulls container images and run them.

PODS: in simple it runs an application.

The cluster architecture: